RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Additional Practice Questions for Chapter S1A: Permutations and Combinations

1 In how many ways can a committee of 3 men and 3 women be chosen from a group of 7 men and 6 women? [2]

The oldest of the 7 men is A and the oldest of the 6 women is B. It is decided that the committee can include at most one of A and B. In how many ways can the committee now be chosen? [3]

[700, 550]

2 A box contains 9 balls. Out of these 9 balls, there are 3 identical red balls, 2 identical yellow balls and 4 numbered green balls (each labelled with a different number from 1 to 4). 3 balls are to be picked out of the box, and the order in which they are picked out does not matter. Find the number of possible selections of 3 balls. [3]

[31]

3 How many 6-digit numbers

(i) are even?

(ii) begin and end with different digits?

[**(i)** 450000, **(ii)** 810000]

4 Find the number of distinct arrangements of the letters of the word '*THERMOMETER*'

(i) if at least 2 '*E*'s are together,

(ii) which must start and end with '*T*' or '*R*'.

[**(i)** 408240, **(ii)** 90720]

5 How many 4-digit numbers greater than 5000 can be formed from the digits 0, 1, 2, 3, 4, 5 if

(i) no digit may be repeated?

(ii) any digit may be repeated?

(ii) only the digit 4 may be repeated?

[**(i)** 60, **(ii)** 215, **(iii)** 73]

6 A school is asked to send a delegation of six pupils selected from six badminton players, six tennis players and five squash players. No pupil plays more than one game. The delegation is to consist of at least one, and not more than three, players drawn from each sport. Giving full details of your working, find the number of ways in which the delegation can be selected.

[9450]

7 Seven cards each have a single digit written on them. The digits on the seven cards are 2, 2, 5, 7, 7, 7, 7 respectively. Find the number of different 4-digit numbers that can be formed by placing four of the cards side by side. [4]

[39]

8 A rectangular table has 7 secured seats, 4 being on one side facing the window and 3 being on the opposite side. In how many ways can 7 people be seated at the table

(i) if 3 people, X and Y and Z must sit on the side facing the window?

(ii) if 2 people, P and Q must sit on opposite sides?

[**(i)** 576, **(ii)** 2880]

9 4 boys, 4 girls and a teacher are to be seated at a round table. How many ways can they be arranged if

(i) there is no restriction?

(ii) the teacher is to be seated between any 2 girls?

(iii) none of the boys are to be seated together?

[**(i)** 40320, **(ii)** 8640, **(iii)** 2880]

10 Three couples who each have a child are to be seated at a round table with ten secured seats. Find the number of ways the nine people can be seated if

(i) they do not mind who they are sitting with, [1]

(ii) none of the children are to be seated on adjacent seats. [3]

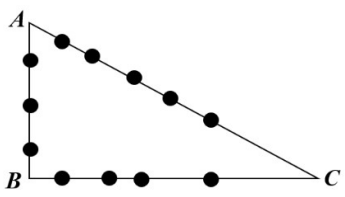
Mr Bean, who knows the three families well, is invited to take a seat at the table. Find the number of ways to seat the ten people if each child is to be seated adjacent to both his or her parents. [3]

[**(i)** 362880; **(ii)** 151200, 48]

11 The following diagram shows 12 **distinct** points on the sides of a triangle ABC.

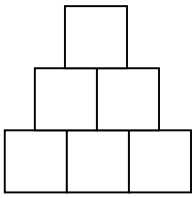
(i) How many line segments are there joining any two points on different sides?

(ii) How many triangles can be formed by joining any one point on side AB, any one point on side BC and any one point on side AC?



[**(i)** 47, **(ii)** 60]

12 Six identical boxes are arranged in 3 rows as shown in the following diagram.



Sandra is given 1 green, 2 blue and 3 red balls. The balls are identical except for their colour. She is to put one ball in each box. Find the number of ways she can do this when

(i) there is no restriction, [2]

(ii) the balls in the bottom row are of different colours, [2]

(iii) there are at least 2 red balls in the bottom row. [3]

[**(i)** 60, **(ii)** 18, **(iii)** 30]

13 There are nine **different** waterslides at an amusement park. A person is allowed to go on each slide once only. However, he can choose to skip the slide or go on it.

(i) If John goes on at least two slides at the amusement park, how many ways can he select the slides to go on?

(ii) At another theme park, there are n different slides. John visits the theme park frequently and selects 2 slides to go on during every visit. At each visit, he makes a different selection, and first realizes that this was no longer possible on the 29th visit. Determine algebraically, the value of n.

(iii) Give an example of a situation involving waterslides to which the expression

$$\frac{n!}{r!(n-r)!} \text{ is the solution.}$$

[**(i)** 502, **(ii)** $n=8$]

14 Ten chefs, six males and four females, qualify for the final phase of a 'Top Chef' competition consisting of nine 'Elimination Challenges'. One chef is eliminated after every challenge and the last chef remaining is the winner of the competition.

(i) In how many ways can the eliminations be done? [1]

(ii) For the first challenge, the ten chefs are divided into two groups of five. In how many ways can the two groups be formed such that each group has at least one female chef? [2]

(iii) After the first challenge, a particular male chef was eliminated. At a photography session before the next challenge, nine chairs are arranged in two rows: four in front and five at the back. In how many ways can the remaining chefs be arranged so that the male chefs and the female chefs must alternate? [3]

[**(i)** 3628800, **(ii)** 120, **(iii)** 5760]

15 2 men and 5 women go to a restaurant. They choose an outdoor round table with 7 seats for their meal. Find the number of ways the group can be seated if

(i) the two men are not seated next to each other. [2]

(ii) one of the women, Mary, is to be seated between the two men. [2]

Before their orders arrive, they request to shift to a table in the 'non-smoking' section of the restaurant. They are then given a round table with 10 seats.
Find the number of ways they can be seated if

(iii) the empty seats are adjacent to each other. [2]

(iv) none of the empty seats are adjacent to each other and there must be more than 1 person between any two empty seats. [2]

[**(i)** 480, **(ii)** 48, **(iii)** 5040, **(iv)** 5040]

16 **(a)** A student is arranging 4 blue flags, 4 red flags, 1 green, 1 yellow and 1 purple flag in a line. All the flags are identical except for the colour. Find the number of different possible ways to arrange the flags if

(i) the green, yellow and purple flags must be placed together, [2]

(ii) no blue flags are placed next to each other, [2]

(iii) a red flag at the beginning and another red flag at the end of the line. [2]

(b) Ten people at a company dinner consist of 7 guests and 3 hosts. The group is to be seated at a round table.

(i) Find the number of arrangements such that there are at least two guests seated between any two hosts. [3]

(ii) Two particular guests could not make it to the dinner. Find the number of ways to arrange the remaining guests and hosts with 10 identical chairs. [3]

[**(a)(i)** 3780, **(a)(ii)** 14700, **(a)(iii)** 7560, **(b)(i)** 30240, **(b)(ii)**181440]

THE END